

Does Syntactic Structure Make Language Models More Robust to Noise?

LSTM vs. RNN Subject-Verb Agreement Under Training-Data Corruption

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Outline

- 1 Motivation
- 2 Background
- 3 Methods
- 4 Results
- 5 Discussion

Motivation: Can grammar survive noisy input?

- Children acquire robust grammatical systems despite hearing noisy and imperfect language
- This observation motivates the classic **Poverty of the Stimulus** debate:
Is grammar learned from data, or supported by innate structure?
- Human experiments cannot systematically manipulate children's linguistic input
- Neural language models provide a controllable testbed:
 - architecture = innate structure
 - training corpus = linguistic environment

RQ1: Can grammar be learned from noisy input?

- How much grammatical corruption can a model tolerate?
- Is there a critical noise threshold beyond which acquisition breaks down?

RQ2: How does noise change generalization?

- Does the model maintain a hierarchical syntactic rule?
- Or does it fall back on a simpler linear nearest-noun heuristic?

RQ3: Does structure buy robustness?

- Compare LSTM and RNNG under identical noise conditions.
- Does explicit syntactic bias help preserve grammatical learning?

Subject-verb agreement: hierarchical vs. linear

- **Hierarchical rule:** agree with the true syntactic subject.
- **Linear rule:** agree with the noun closest to the verb.

Example

The keys to the cabinet _____ on the table.

- | | |
|--|---|
| • True subject = <i>keys</i> (plural) | • Hierarchical $\Rightarrow$ are |
| • Nearest noun = <i>cabinet</i> (singular) | • Linear $\Rightarrow$ is |

Key idea: SVA with attractors creates cases where the two rules make *different* predictions. This is our diagnostic lever.

Attractors as the diagnostic case

- An **attractor** is an intervening noun that mismatches the subject in number (e.g. *cabinet* between *keys* and the verb).
- On **no-attractor** items, both rules agree $\Rightarrow$ accuracy is high for either strategy.
- On **attractor** items, only the hierarchical rule is correct $\Rightarrow$ success requires ignoring the misleading linear cue.
- The **attractor gap** (no-attractor acc — attractor acc) measures how much the model relies on the linear shortcut.

Two competing learners

LSTM

- Sequential recurrent model
- No explicit syntactic structure
- Learns grammar from statistical regularities

Weak structural prior

RNNG

- Generates sentences and parse trees jointly
- Explicit hierarchical representations
- Strong syntactic inductive bias

Strong structural prior

Key question: Does explicit syntactic structure help preserve hierarchical rules under noisy input?

SVA as a syntax benchmark

- Linzen et al. (2016): LSTMs learn agreement but struggle with attractors.
- Gulordava et al. (2018): evidence for abstract syntactic generalization.

Structural inductive bias

- Dyer et al. (2016): Recurrent Neural Network Grammar (RNNG).
- Kuncoro et al. (2018): explicit syntax improves long-distance dependencies.

Research gap

- Prior work tests models on clean data; little asks whether grammatical rules can still be *acquired* under noisy supervision.
- No systematic study of how SVA corruption affects rule acquisition, hierarchical vs. linear generalization, or LSTM vs. RNNG robustness.

Noise injection: five corrupted corpora

- Base: Linzen et al. (2016) SVA corpus, $\sim 1.58\text{M}$ Wikipedia sentences with vocab capped at the 10K most frequent types.
- Each sentence annotated with subject, verb, verb position, and $\#$ intervening attractor nouns.
- Build five training corpora, identical except agreement noise: flip the verb's number with probability p .

$$p \in \{0, 0.004, 0.02, 0.10, 0.50\} \quad (\text{baseline} \rightarrow \text{high})$$

- Flip via deterministic inflection (lemminflect + irregular copulas/auxiliaries); all other lexical/syntactic content preserved.
- Training-set size held constant $\Rightarrow$ isolates noise.

Leak-free evaluation protocol

- The five corpora are row-aligned with the source corpus.
- Reserve one fixed held-out set: 20k sentence indices (seed 1234).
- Those rows excluded from **every** training corpus $\Rightarrow$ no test leakage under any noise condition.
- Test items built from the **uncorrupted** source $\Rightarrow$ all test sentences are grammatical.
- Same test set for all conditions. Only the training noise varies.

Minimal-pair test

- For each held-out sentence: a prefix, the grammatical verb v^+ , and its number-flipped counterpart v^- .
- Model is **correct** iff it prefers the grammatical form:

$$P(v^+ \mid \text{prefix}) > P(v^- \mid \text{prefix})$$

- Each pair tagged `has_attractor`: an intervening noun mismatching the subject in number.

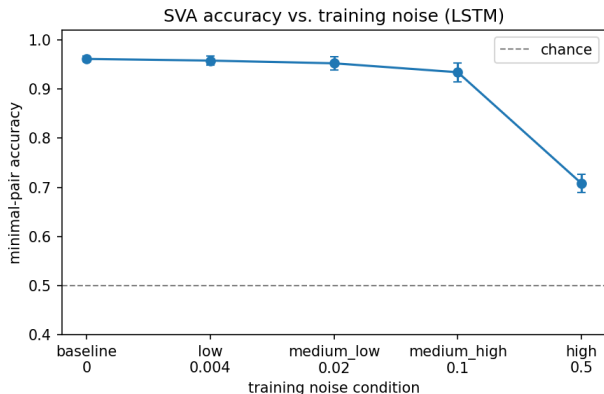
- **Sequential baseline:** word-level LSTM LM (Linzen 2016 / Gulordava 2018 style). 2 layers, tied embeddings, dropout. Trained from scratch per condition.
- **Structurally-biased:** RNNG (Dyer et al. 2016), which builds explicit hierarchical structure.
- Core question: does a structural bias make hierarchical agreement *more robust* to noisy supervision?
- Fallback: ON-LSTM (Shen et al. 2019), which has a softer structural bias and doesn't need parse supervision.

- Primary: minimal-pair **accuracy** (grammatical $>$ ungrammatical).
- Secondary (graded): **surprisal** of the correct verb, $s = -\log_2 P(v^+ \mid \text{prefix})$.
- Split accuracy by attractor presence; reliance on the linear cue = **attractor gap**

$$\Delta = \text{acc}_{\text{no-attr}} - \text{acc}_{\text{attr}}$$

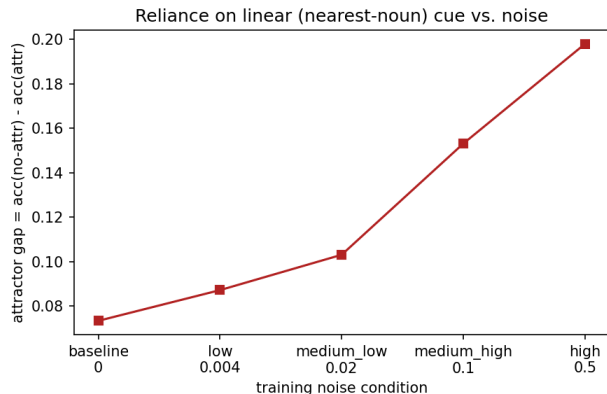
- Small $\Delta \rightarrow$ hierarchical rule; large $\Delta \rightarrow$ nearest-noun (linear) rule.
- Averaged over seeds $\{1, 2, 3\}$; central analysis: Δ vs. noise.

Accuracy holds up until 50% noise



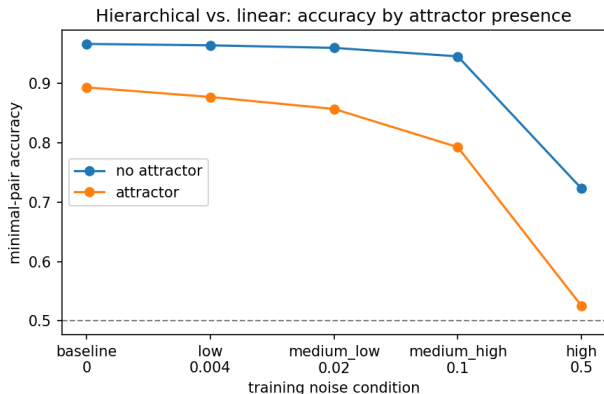
- Stays flat up to 10% noise (0.961 $\rightarrow$ 0.934)
- Drops sharply at 50%, down to 0.708
- The model can handle about 1 bad verb in 10 before it breaks
- Not a gradual decline, more like a threshold

The attractor gap grows with noise



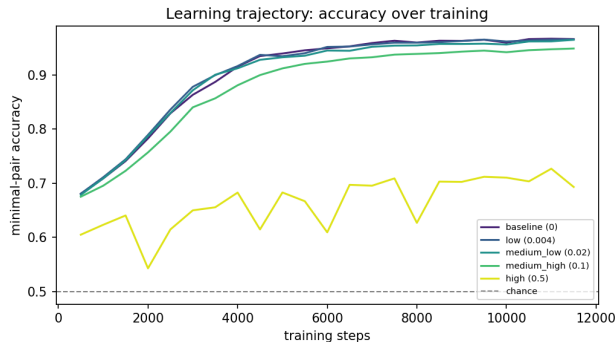
- Gap = $\text{acc}(\text{no attractor}) - \text{acc}(\text{attractor})$
- Goes from 0.073 to 0.198 as noise increases
- Tells us how much the model leans on the nearest-noun shortcut
- More noise means more reliance on the linear rule

Attractor items are where it falls apart



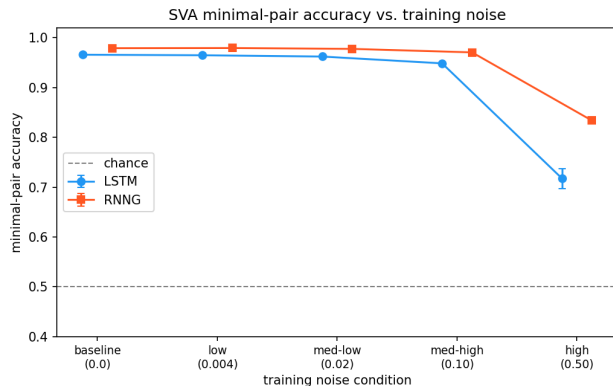
- Without attractors, accuracy stays above 0.94 until 50%
- With attractors, it drops to near chance (0.525)
- Attractor items are the only cases where the two rules disagree
- So that's exactly where the model struggles first

Learning over time: delay vs. prevention



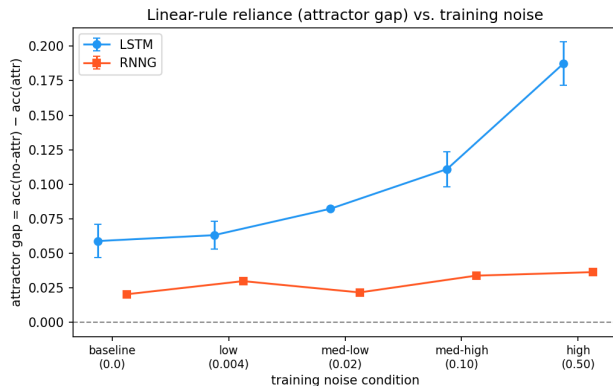
- Up to 10% noise: reaches the same accuracy, just takes longer (**delay**)
- At 50%: plateaus around 0.69 and never catches up (**prevention**)
- Clean model: gap shrinks over training (0.15 $\rightarrow$ 0.05)
- 50% model: gap actually *grows* (0.14 $\rightarrow$ 0.20)

Does structure help? RNNG vs. LSTM



- RNNG = LSTM plus an explicit syntactic (tree) bias
- More accurate than the LSTM at *every* noise level
- Gap widens where noise bites hardest: at 50%, 0.834 vs. 0.708
- Same held-out set ($\sim 20k$ pairs); single full-scale seed so far

RNNG keeps the hierarchical rule under noise



- LSTM's attractor gap grows $2.7\times$ with noise ($0.073 \rightarrow 0.198$)
- RNNG's stays low and flat ($0.020 \rightarrow 0.036$)
- At 50% noise the RNNG gap is *half* the LSTM's clean-data gap
- Structural bias preserves the hierarchical rule where the LSTM defaults to the linear shortcut

When does the rule break?

- **It's a threshold, not a gradual decline.** The model absorbs noise fine up to 10%, then snaps at 50%. Similar to how a child's grammar survives occasional speech errors around them.
- **The attractor gap catches it early.** It climbs from 0.07 to 0.20 even while overall accuracy still looks fine.
- **Low noise delays learning, high noise prevents it.** At low noise the model still gets there eventually (the curves converge). At 50% it plateaus and never learns the rule.

Does structure buy robustness?

- The big question: does a model with built-in syntax (RNNG) hold up better than a plain LSTM? RNNG training is still in progress.
- **What we know from the LSTM:** it picks up the hierarchical rule from clean data, but drifts toward the nearest-noun shortcut as noise grows.
- If RNNG keeps the hierarchical rule where the LSTM gave up, that points to structure helping with robustness.
- If both models break at the same point, then maybe it's really about the quality of the input, not the architecture. That ties back to the poverty-of-the-stimulus question.

What we found

- LSTMs can handle moderate SVA noise, but there's a sharp breaking point.
- Noise doesn't just make the model worse overall. It changes *which rule* the model learns, pushing it toward the linear nearest-noun shortcut.
- At 50% noise, more training actually makes things worse, not better.

Limitations and next steps

- Our noise is synthetic and uniform; the corpus is templatic; trajectories are single-seed.
- Finish the RNNG comparison; try ON-LSTM as a middle ground.
- Future: more naturalistic noise patterns and larger-scale training.

Thank you!

Questions?